

OCR Guide

Enterprise AI Knowledge Assistant — Day 37 Reference Document

Purpose

This sample document is designed for the Day 37 citation-generation test. It contains clear headings and page-oriented content so retrieved chunks can be associated with document, page, and chunk information.

1. What is OCR?

Optical Character Recognition (OCR) converts text contained in scanned pages or images into machine-readable text. In an enterprise knowledge assistant, OCR can make scanned PDFs searchable and available for downstream cleaning, chunking, embedding, and retrieval.

2. OCR Processing Pipeline

A typical pipeline is: scanned document → image extraction → OCR engine → raw text → text cleaning → chunking → embeddings → vector database → semantic search.

3. OCR Quality

OCR accuracy can be affected by image resolution, font style, page rotation, skew, background noise, tables, handwriting, and low contrast. Preprocessing may improve the quality of extracted text.

4. Text Cleaning After OCR

OCR output may contain unnecessary line breaks, repeated spaces, broken words, unusual characters, and page artifacts. A cleaning stage can normalize whitespace, remove unwanted characters, and prepare text for chunking.

5. Chunking

Chunking divides a long document into smaller text units. Chunk size and overlap should be selected according to retrieval requirements. Each chunk should retain enough context to represent its topic meaningfully.

6. Embeddings

An embedding model converts each chunk into a numerical vector. Semantically related chunks tend to have vectors that are closer according to the selected similarity metric.

7. FAISS Vector Search

FAISS can store embeddings and retrieve vectors that are nearest to a query embedding. The returned vector ID must remain aligned with the corresponding metadata record.

8. Metadata

Metadata connects a retrieved vector to useful source information. Typical fields include document name, page number, chunk ID, source, character indexes, and chunk text.

9. Citation Generation

A citation service converts raw retrieval information into a compact citation that can be shown in the frontend. For example, document, page, and chunk fields can be formatted as:

[OCR_Guide.pdf | Page 4 | Chunk chunk_12]

The citation should preserve the source identity so users can understand where a retrieved piece of information originated.

10. Example Retrieved Result

Field	Example
Document	OCR_Guide.pdf
Page	4
Chunk	chunk_12
Distance	0.2381

11. Day 37 Test Objective

Retrieve multiple chunks, pass retrieval results to a citation service, generate citations for every result, and display those citations in the frontend.

12. Expected Citation Flow

FAISS search → vector IDs → MetadataStore → retrieved result dictionaries → CitationService → formatted citations → Streamlit display.

13. Citation Test Checklist

1. Upload OCR_Guide.pdf.
2. Create embeddings.
3. Run similarity search.
4. Confirm document, page, and chunk metadata.
5. Pass search results to CitationService.
6. Confirm the citation string is generated.
7. Display the citation below each retrieved chunk.